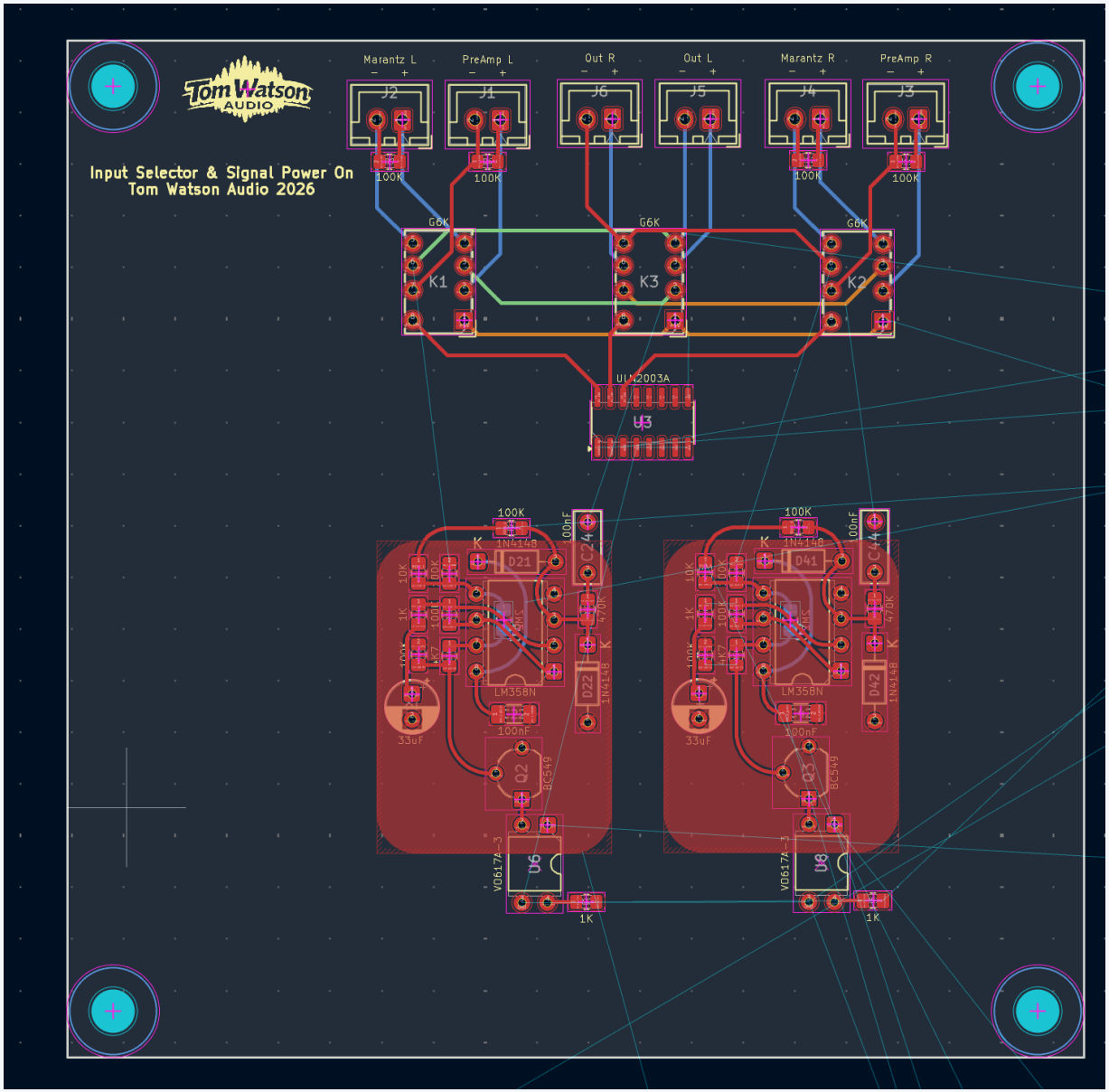
Your starting layout: what to keep, what to improve

Reference: your supplied screenshot. Colours in KiCad do not establish a copper zone's electrical net.



KEEP: audio along the top

Connectors and K1/K2/K3 belong together. Keep each signal beside its own return; minimise the through path and the area enclosed by that pair.

KEEP: two compact detector blocks

Place each detector below its selector. A short sensing branch is useful, but keeping the high-impedance node at LM358 pin 3 tiny matters more than exact left/right symmetry.

IMPROVE: coil-current corridor

U3 below the relays is sensible. Route coil supply, switched coil leads and their control return below/around the contact wiring. Avoid long parallel runs beside audio traces.

CHECK: local decoupling

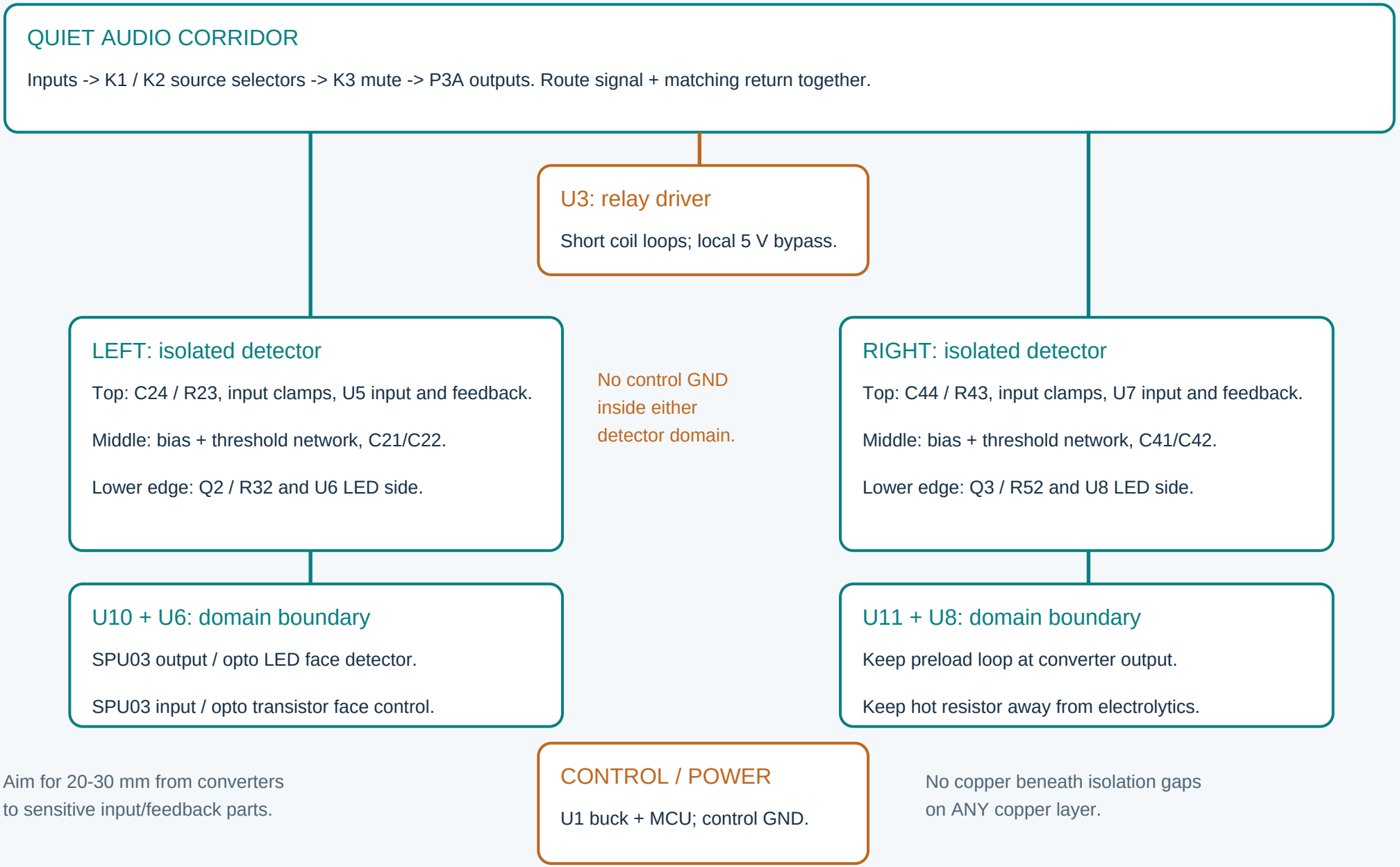
The 100 nF parts below the LM358s deserve attention: locate C22/C42 at pin 8, with a short return to pin 4/local return copper. Component proximity alone is not enough.

RESERVE: lower edge for power

Keep the buck, SPU03 modules and preload resistors out of the audio corridor. Let the converters straddle the boundary between control and isolated detector domains.

Recommended floorplan

Conceptual arrangement, not to scale. Space targets are practical starting points, not guaranteed noise limits.



Placement details that matter most

Component references follow the existing detector design. Confirm nets in KiCad before routing.

1 Make the sensitive nodes small

U5/U7 pin 3: cluster R23/R43 (470k), R24/R44 (100k) and D21/D22 or D41/D42 at the input pin. Keep C24/C44 nearby. Route the longer sensing connection BEFORE the 470k resistor, not from its high-impedance output. Keep pin 2 feedback R25/R45 close to pins 1 and 2; place R26/R46 and C25/C45 nearby.

2 Decouple at the actual supply pins

C22/C42 (100n): aim for a few millimetres from pin 8, with a short, broad local-return path to pin 4. A backside capacitor under the DIP can help. R21/R41 + C21/C41 belong at the quiet-supply entry to each detector. C20/C40 belong directly across the SPU03 INPUT supply pins.

3 Separate quiet references from pulsed currents

Keep bias dividers R22/R34 and R42/R54 with C23/C43 near the op-amp. Keep threshold/hysteresis resistors R27-R29 and R47-R49 near pins 5/7. Put Q2/Q3, LED resistors R32/R52 and optocouplers together at the lower edge. Return their current to the raw isolated supply locally, not through bias/input return traces.

4 Use deliberate copper domains

Each detector needs a contiguous local-return area: SEL_L_RETURN or SEL_R_RETURN. Control GND must not pour under either detector or across SPU03/opto isolation boundaries. Do not join L/R on this board merely because the P3A joins them externally. Take the audio-return sensing branch from the selected return; keep preload and LED currents out of the main audio-return route.

5 Keep heat and switching loops local

Place each preload beside its SPU03 output, with short outgoing/return paths. Aim for at least 10 mm body-to-body from electrolytics and raise axial power resistors about 5 mm where enclosure height permits. Keep U1 and its input/output capacitors compact. Feed U3 coils separately from detector-module feeds; clamp COM returns to the coil supply locally.

6 Route and verify in this order

Audio signal/return pairs first; detector inputs/feedback second; local decoupling and isolated power loops third; coil/control wiring last. Avoid control traces through detector islands. Where crossings are unavoidable, use short near-perpendicular crossings on different layers. Check zone net assignments and all-layer isolation, then refill and run DRC.

Scope: screenshot-based placement advice, with reference to project detector circuitry; saved PCB positions differ from the screenshot. No routing, connectivity, thermal or acoustic sign-off. Placement does not resolve the existing ULN2003A/5 V relay pickup-margin issue.

References: TI LM358 datasheet, layout section; RECOM R-78K-2.0 and Mean Well SPU03 datasheets.